Silicon Photonics Course Project: Design Document

Multi-Channel Mach-Zehnder Interferometer (MZI) Spectrum Analysis Subsystem

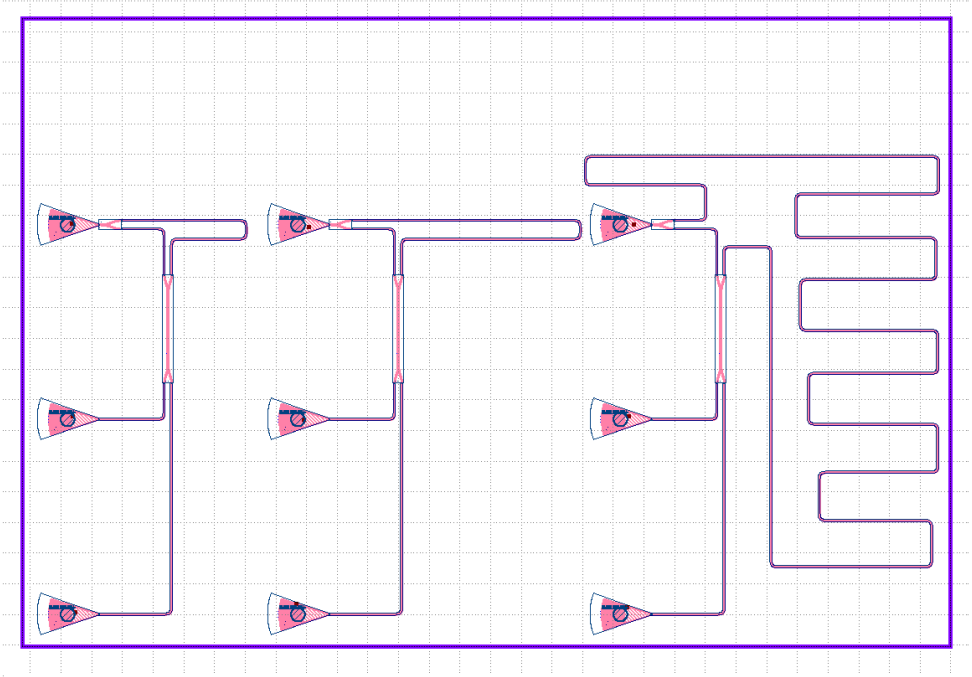
1. General Information & Layout Specifications

This design document outlines the layout parameters, photonic architecture, and spatial constraints for the assigned Silicon Photonics Multi-Project Wafer (MPW) course project. The circuit is designed for automated optical probing using standard single-mode fiber arrays on the SiEPIC EBeam platform.

Parameter / Requirement	Specification Details
Target Platform & PDK	SiEPIC EBeam PDK (220 nm SOI Process, 2-3 µm PECVD SiO2 Cladding)
Assigned Layout Footprint	605 µm (Width) × 410 µm (Height) Floorplan Rectangle (GDS Layer #99)
Polarization & Optical Mode	Transverse Electric (TE_0) mode at 1550 nm operating wavelength
Grating Coupler Array Pitch	127 µm vertical center-to-center pitch, all couplers right-facing

2. Circuit Architecture: 3× Mach-Zehnder Interferometers

The core photonic architecture consists of three distinct Mach-Zehnder Interferometers (MZIs) integrated with unbalanced arm lengths. The objective is to evaluate Free Spectral Range (FSR) scaling, extinction ratios, and phase stability across varying path differential delays (ΔL).



3. Automated Probing Compliance & Text Labeling

To interface with the automated motorized fiber array measurement station at the University of Washington / SiEPIC testing facility, all optical ports strictly adhere to standard positioning and GDS text label conventions:

• **Fiber Array Port Alignment:** Fiber 2 serves as the primary optical input connected to the tunable laser source. Fibers 1, 3, and 4 are assigned to output photodetectors. Maximum alignment power fine-tuning is performed automatically using Fibers 1 and 3.

• **Text Label Rules (GDS Layer #10):** Labels must follow strict case-sensitive formatting (e.g., opt_in_TE_1550_device_MZI_VaryingArms) positioned exactly at the tip of the input grating coupler. No additional underscores are permitted.

4. Spatial Floorplan Analysis & Trade-offs

FLOORPLAN & CALIBRATION TRADE-OFF SUMMARY

- Top Free Space Utilization Attempt: Initial layout plans attempted to utilize the vacant upper region of the 605 µm × 410 µm cell by adding dedicated reference calibration circuits (e.g., straight baseline waveguides / loopback structures).
- Vertical Clearance Bottleneck: Due to the fixed height constraint (410 µm) combined with the required 127 µm vertical pitch between grating couplers, there was insufficient vertical height available to route and place additional grating couplers.
- Design Resolution: Standalone calibration circuits were omitted to avoid violating minimum waveguide spacing and bend radius design rules. Spatial resources were dedicated exclusively to optimizing the 3 MZI devices for maximum performance and reliable automated probing.